Harare Institute of Technology

SCHOOL OF INFORMATION SCIENCE AND TECHNOLOGY

HISA

ISS 2102: Number Theory

Assignment 1

DUE : 16:30 HOURS **26 September 2025**

INSTRUCTIONS TO CANDIDATES

- Attempt **ALL** questions.
- Set m to be the last three digits of your registration number.
- Export your documents to PDF then use the link provided to submit.
- Late submissions will not be accepted.
- The Lecturer reserves the right to leave ungraded any assignment that is disorganized, untidy or incoherent.

1. Find: [4]

   (a) $\left\lfloor \frac{m}{2024} \right\rfloor$. (b) $\left\lceil \frac{2024}{m} \right\rceil$.

2. Use the division algorithm to find the quotient and remainder when -1002 is divided by m. [2]

3. Use the Euclidian algorithm to find $\gcd(m, 2023)$. [2]

4. What is the lowest common multiple of m and 2024? [2]

5. Is m relatively prime to 2023? [2]

6. Find all prime numbers less than 160. [2]

7. Find all prime numbers between 23 and 570. [2]

8. Find the prime factors of m. [2]

9. Find the number of prime factors of $280m$. [2]

10. Find the sum of the exponents in the prime decomposition of m. [2]

11. Find the prime factors of m. [2]